

We now explain how many concepts from the theory of schemes may be applied to algebraic stacks.

Let $\mathcal{P}$ be a property of morphisms of schemes, stable by étale change of base, and of a local nature (for the étale topology) on the target.

For instance: being an open immersion with dense image, being dominant, birational...

A representable morphism of algebraic stacks $f : T_1 \rightarrow T_2$ is said to have property $\mathcal{P}$ if for one (and hence for every) surjective étale morphism $x : X \rightarrow T_2$, the morphism of schemes deduced by base change

$$f' : X \times_{T_2} T_1 \longrightarrow X$$

has that property.

Let $\mathcal{P}$ be a property of morphisms of schemes, which, at source and target, is of a local nature for the étale topology. This means that, for any family of commutative squares

$$\begin{array}{ccc} X_i & \xrightarrow{g_i} & X \\ f_i \downarrow & & \downarrow f \\ Y_i & \xrightarrow{h_i} & Y \end{array}$$

where the g_i (resp. h_i) are étale and cover X (resp. Y):

$$\mathcal{P}(f) \iff \forall i \mathcal{P}(f_i).$$

For instance: f flat, smooth, étale, unramified, normal, locally of finite type, locally of finite presentation.

If $f : T_1 \rightarrow T_2$ is a morphism of algebraic stacks, we say f has property $\mathcal{P}$ if for one, then necessarily for every, commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & T_1 \\ f' \downarrow & & \downarrow f \\ Y & \xrightarrow{y} & T_2 \end{array}$$

where X and Y are schemes and x, y are étale and surjective, f' has property $\mathcal{P}$.

Similarly, if $\mathcal{P}$ is a property of schemes, of a local nature for the étale topology, an algebraic stack T will be said to have property $\mathcal{P}$ if for one (and hence for every) surjective étale morphism $x : X \rightarrow T$, X has property $\mathcal{P}$. This applies to, for instance, the properties of being regular, normal, locally noetherian, of characteristic p , reduced, Cohen–Macaulay...

An algebraic stack T will be called *quasi-compact* if there exists a surjective étale morphism $x : X \rightarrow T$ with X quasi-compact. A morphism $f : T_1 \rightarrow T_2$ of algebraic stacks will be called *quasi-compact* if for any quasi-compact scheme X over T_2 , the fibre product $T_1 \times_{T_2} X$ is quasi-compact. It is enough to test the condition for a surjective family $f_i : X_i \rightarrow T_2$. We define a morphism $f : T_1 \rightarrow T_2$ to be of *finite type*, if it is quasi-compact, and locally of finite type; of *finite presentation*, if it is quasi-compact, quasi-separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a “proper morphism” and the analogue of Chow’s lemma.

A morphism $f : T_1 \rightarrow T_2$ is said to have a property $\mathcal{P}$, locally on T_2 , if there exists a surjective étale morphism $x : X \rightarrow T_2$ such that the morphism f' deduced from f by the change of base by x has property $\mathcal{P}$.

Definition (4.11). — A morphism $f : T_1 \rightarrow T_2$ is proper if it is separated, of finite type and if, locally over T_2 , there exists a commutative diagram

$$\begin{array}{ccc} T_3 & \xrightarrow{g} & T_1 \\ & \searrow h & \swarrow f \\ & & T_2 \end{array}$$

with g surjective and h representable and proper.

The following form of Chow’s lemma will be sufficient for our purposes.

Theorem (4.12). — Let S be a noetherian scheme and f be a morphism from an etale site T to S . We assume f to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{array}{ccccc} & & g & & \\ & & \swarrow & & \searrow \\ T & \xleftarrow{\quad} & T' & \xrightarrow{\quad} & T'' \\ & \searrow & \downarrow f' & \swarrow f'' & \\ & & S & & \end{array}$$

in which T' and T'' are schemes and such that

- (i) g is proper, surjective and generically finite;
- (ii) j is an open immersion;
- (iii) f'' is projective.

Using (4.12), it is easy to extend the cohomological theory of coherent sheaves to the present situation. In fact if $f : T_1 \rightarrow T_2$ is a proper morphism of noetherian algebraic stacks and if $\mathcal{F}$ is a coherent sheaf on T_1 , then the $R^i f_*(\mathcal{F})$ are coherent sheaves on T_2 .

However, the $R^i f_*(\mathcal{F})$ don't need to be zero for i large enough. Let S be a scheme and G a finite group. We denote by p the projection $p : [S/G] \rightarrow S$. Quasi-coherent (resp. coherent) sheaves of modules on $[S/G]$ may (and will) be identified with quasi-coherent (resp. coherent) sheaves of modules on S , on which G acts. One has

$$R^i p_*(\mathcal{F}) \simeq H^i(G, \mathcal{F}).$$

In general, the (quasi-coherent sheaf) cohomology of algebraic stacks appears as a mixture of finite group cohomology and of scheme cohomology.

The *disjoint sum* T of a family $(T_i)_{i \in I}$ of stacks is the stack a section of which over a scheme X consists of

- (i) a decomposition $X = \coprod_i X_i$ of X ;
- (ii) a section of T_i over X_i for each i .

The *void stack* $\emptyset$ is the one represented by the void scheme.

A stack is *connected* if it is non-void and is not the disjoint sum of two non-void stacks.

Proposition (4.13). — A locally noetherian algebraic stack is in one and only one way the disjoint sum of a family of connected algebraic stacks (called its connected components).

We denote by $\pi_0(T)$ the set of connected components of a locally noetherian algebraic stack T . If $x : X \rightarrow T$ is surjective and etale, $\pi_0(T)$ is the cokernel of the two maps

$$\pi_0(X \times_T X) \rightrightarrows \pi_0(X) \longrightarrow \pi_0(T).$$

Proposition (4.14). — Let T be an algebraic stack of finite type over a field k . Then, T is connected if and only if there exists a connected scheme X , of finite type over k , and a surjective morphism from X to T .

An open subset U of an algebraic stack T is a full subcategory $U \subset T$ which is an algebraic stack, which contains together with any $t \in \text{Ob}(T)$ all isomorphic t' and such that the inclusion $j : U \rightarrow T$ is representable by open immersions. The open subsets of T correspond bijectively to the open subsets of its etale site.

For each open subset U of T , there exists one and only one full subcategory $T - U$ of T , which is an algebraic stack, which contains together with any $t \in \text{Ob}(T)$ all isomorphic t' and such that

- (i) $T - U$ is reduced;
- (ii) the inclusion map $i : T - U \rightarrow T$ is representable by closed immersions;
- (iii) for any etale surjective morphism $x : X \rightarrow T$, the inverse image of $T - U$ on X is the complement of the inverse image of U .

An algebraic stack F in T satisfying (i) and (ii) is a *closed subset* of T , and the functor $U \mapsto T - U$ is an isomorphism of the set of open and the set of closed subsets of T . If F satisfies only (ii), F_{red} satisfies (i) and (ii) so that F defines a closed subset of T .

An algebraic stack T is *irreducible* if it is not the union of two closed subsets, non-void and distinct from T .

Proposition (4.15). — A noetherian algebraic stack T is in one and only one way the union of irreducible closed subsets, none of which contains any other. They are called the irreducible components of T . If U is an open dense subset of T , the irreducible components of U are the non-void intersections of U with the irreducible components of T .

Each irreducible component of T is contained in a connected component of T . Conversely:

Proposition (4.16). — The connected components of a normal noetherian algebraic stack are irreducible.

Theorem (4.17). — Let f be a morphism of finite type from an algebraic stack T to a noetherian scheme S . For $s \in S$, let $n(s)$ be the number of connected components of the geometric fibre of T at s . Then

- (i) $n(s)$ is a constructible function of s ;
- (ii) if f is proper and flat, then $n(s)$ is lower-semi-continuous;
- (iii) if f is proper flat, and has geometrically normal fibres, then $n(s)$ is constant.

Let $f : T \rightarrow S$ be a morphism of finite type from an algebraic stack T to a noetherian scheme S . Assume that the diagonal map $T \rightarrow T \times_S T$ is separated and quasi-compact.

Theorem (4.18) (Valuative criterion for separation). — The morphism f is separated if and only if, for any complete discrete valuation ring with algebraically closed residue field and any commutative diagram

$$\begin{array}{ccc} & T & \\ g_1 \nearrow & & \searrow f \\ \text{Spec}(V) & \longrightarrow & S \end{array}$$

any isomorphism between the restrictions of g_1 and g_2 to the generic point of $\text{Spec}(V)$ can be extended to an isomorphism between g_1 and g_2 .

This criterion is nothing other but the valuative criterion of properness (EGA, II, 7.3.8) applied to the (representable) diagonal morphism.

Theorem (4.19) (Valuative criterion for properness). — If f is separated, then f is proper if and only if, for any discrete valuation ring V with field of fractions K and any commutative diagram

$$\begin{array}{ccccc} & & T & & \\ & g \nearrow & & \searrow f & \\ \text{Spec}(K) & \longrightarrow & \text{Spec}(V) & \longrightarrow & S \end{array}$$

there exists a finite extension K' of K such that g extends to $\text{Spec}(V')$, where V' is the integral closure of V in K'

$$\begin{array}{ccccc} & & T & & \\ & & \nearrow & \searrow f & \\ \text{Spec}(K') & \longrightarrow & \text{Spec}(V') & & \\ \downarrow & \nearrow & \downarrow & & \\ \text{Spec}(K) & \longrightarrow & \text{Spec}(V) & \longrightarrow & S \end{array}$$

To prove a given f is proper, it suffices to verify the above criterion under the additional hypothesis that V is complete and has an algebraically closed residue field. Further, given a dense open subset U of T , it is enough to test only g 's which factor through U .

Proposition (4.20). — Let $\mathcal{S}$ be an algebraic stack. The functor which, to any algebraic stack over $\mathcal{S}$, $f : \mathcal{T} \rightarrow \mathcal{S}$, associates the $\mathcal{O}_{\mathcal{S}}$ -sheaf of algebras $f_ \mathcal{O}_{\mathcal{T}}$ induces an equivalence of categories between:*

- (i) *the category of algebraic stacks representable and affine over $\mathcal{S}$;*
- (ii) *the dual of the category of quasi-coherent $\mathcal{O}_{\mathcal{S}}$ -algebras.*

Let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_{\mathcal{S}}$ -algebras on an algebraic stack $\mathcal{S}$. For each étale morphism $x : U \rightarrow \mathcal{S}$, with U affine, let $\mathcal{A}'(U)$ be the integral closure of $\Gamma(U, \mathcal{O}_{\mathcal{S}})$ in $\mathcal{A}(U)$. By (EGA, II, 6.3.4), the $\mathcal{A}'(U)$ for variable U are the sections over U of a quasi-coherent sheaf $\mathcal{A}'$ on $\mathcal{S}$, which will be called the integral closure of $\mathcal{O}_{\mathcal{S}}$ in $\mathcal{A}$.

Let $f : \mathcal{T} \rightarrow \mathcal{S}$ be representable and affine. The algebraic stack which is associated by (4.20) to the integral closure of $\mathcal{O}_{\mathcal{S}}$ in $f_*\mathcal{O}_{\mathcal{T}}$ will be called the normalisation of $\mathcal{S}$ with respect to $\mathcal{T}$. Its formation is compatible with any étale change of basis.

Theorem (4.21). — Let $\mathcal{S}$ be a quasi-separated stack over a noetherian scheme S . Assume that

- (i) *the diagonal map $\mathcal{S} \rightarrow \mathcal{S} \times_S \mathcal{S}$ is representable and unramified;*
- (ii) *there exists a scheme X of finite type over S and a smooth and surjective S -morphism from X to $\mathcal{S}$.*

Then, $\mathcal{S}$ is an algebraic stack of finite type over S .

M. Artin has developed powerful methods to relate pro-representability of a stack to the existence of étale surjective maps $x : X \rightarrow \mathcal{S}$.

§ 5. Second proof of the irreducibility theorem.

Let $\mathcal{M}_g$ ($g \geq 2$) be the stack whose category of sections over a scheme S is the category of stable curves of genus g over S , the morphisms being the isomorphisms of schemes over S . By (1.11), the diagonal morphism

$$\Delta : \mathcal{M}_g \longrightarrow \mathcal{M}_g \times \mathcal{M}_g$$

is representable, finite and unramified.

We saw in § 1 that the stack classifying the tricanonically embedded stable curves of genus g is represented by a scheme H_g , smooth and of finite type over $\text{Spec}(\mathbb{Z})$. The “forgetful” morphism

$$H_g \longrightarrow \mathcal{M}_g$$

is representable, smooth and surjective. Indeed, if $p : C \rightarrow S$ is a stable curve over S , defining a morphism

$$\gamma : S \longrightarrow \mathcal{M}_g,$$

then the fibre product $H_g \times_{\mathcal{M}_g} S$ is the scheme, smooth over S , of isomorphisms between the standard projective space of dimension $5g - 6$ over S and $\mathbf{P}(p_*(\omega_{C/S}^{\otimes 3}))$. We deduce from this and (4.21) that:

Proposition (5.1). — $\mathcal{M}_g$ is a separated algebraic stack of finite type over $\text{Spec}(\mathbb{Z})$.

Let us denote by $\mathcal{M}_g^0$ the open subset of $\mathcal{M}_g$ which “consists of” smooth curves, and by $\mathcal{Z}_g$, the “universal curve” over $\mathcal{M}_g$, the algebraic stack classifying pointed stable curves.

Theorem (5.2). — The algebraic stacks $\mathcal{M}_g$ and $\mathcal{Z}_g$ are proper and smooth over $\text{Spec}(\mathbb{Z})$ and the complement of $\mathcal{M}_g^0$ in $\mathcal{M}_g$ is a divisor with normal crossings relative to $\text{Spec}(\mathbb{Z})$.

Proof. — Let $x : X \rightarrow \mathcal{M}_g$ be étale and consider the following commutative diagram of algebraic stacks of finite type over $\text{Spec}(\mathbb{Z})$:

$$\begin{array}{ccc}
 X \times_{\mathcal{M}_g} \mathcal{Z}_g & \xrightarrow{\quad} & \mathcal{Z}_g \\
 \downarrow & \searrow & \downarrow \\
 X \times_{\mathcal{M}_g} H_g & \xrightarrow{x'} & H_g \\
 \downarrow & & \downarrow q \\
 X \times_{\mathcal{M}_g} \mathcal{Z}_g & \xrightarrow{\quad} & \mathcal{Z}_g \\
 \downarrow & \searrow & \downarrow p \\
 X & \xrightarrow{x} & \mathcal{M}_g
 \end{array}$$

In this diagram, the four horizontal arrows are étale and the four vertical arrows are smooth and surjective. As H_g and $\mathcal{Z}_g$ (p. 78) are smooth over $\text{Spec}(\mathbb{Z})$, so are X and $X \times_{\mathcal{M}_g} \mathcal{Z}_g$. In addition,

$$q'^{-1}x^{-1}(\mathcal{M}_g^0) = x'^{-1}(H_g^0)$$

is the complement of a divisor with normal crossings relative to $\mathrm{Spec}(\mathbb{Z})$, so that the inclusion of $x^{-1}(\mathcal{M}_g^0)$ in X has the same property. This being true for any x , $\mathcal{M}_g$ and $\mathcal{Z}_g$ are smooth over $\mathrm{Spec}(\mathbb{Z})$ and $\mathcal{M}_g^0$ is the complement of a divisor with normal crossings relative to $\mathrm{Spec}(\mathbb{Z})$. In particular $\mathcal{M}_g^0$ is dense in $\mathcal{M}_g$.

We may now use the valuative criterion for properness in its modified form (4.19) to deduce the properness of $\mathcal{M}_g$ from the stable reduction theorem. The properness of $\mathcal{Z}_g$ then results from that of p .

(5.3) Let $p : X \rightarrow S$ be a stable smooth curve of genus $g \geq 2$ over S . If $k \in \mathbb{N}$ is invertible in $\mathcal{O}_S$, the sheaf $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ on S_{et} is locally free over $\mathbb{Z}/k\mathbb{Z}$, of rank $2g$, and the cup-product is a non degenerate alternating form

$$R^1p_*(\mathbb{Z}/k\mathbb{Z}) \otimes R^1p_*(\mathbb{Z}/k\mathbb{Z}) \longrightarrow R^2p_*(\mathbb{Z}/k\mathbb{Z}) \simeq \mu_k^{\otimes -1}.$$

Locally on S_{et} , μ_k is isomorphic to $\mathbb{Z}/k\mathbb{Z}$, and thus, locally, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with a non degenerate symplectic form with values in $\mathbb{Z}/k\mathbb{Z}$, which is determined up to a unit in $\mathbb{Z}/k\mathbb{Z}$, or, as we shall say, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with an homogeneous symplectic structure. The constant sheaf $(\mathbb{Z}/k\mathbb{Z})^{2g}$ will be provided with the homogeneous symplectic structure induced by the standard symplectic structure of $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

Definition (5.4). — A Jacobi structure of level k on X is an isomorphism (respecting their homogeneous symplectic structures) between $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ and $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

(5.5) Given a section s of p , and a set of prime numbers $\mathbf{P}$ including all residue characteristics of S , the specialisation theorem for the fundamental group enables one to construct a pro-object $\pi_1(X/S, s)^{(\mathbf{P})}$ of the category (l.c. $\mathrm{gr}^{(\mathbf{P})}$) of locally constant sheaves of finite groups of order prime to $\mathbf{P}$, with the properties

- (i) $\mathrm{Hom}_S(\pi_1(X/S, s)^{(\mathbf{P})}, G) = R^1p_*(X \bmod s, p^*G) = R^1p_*(\mathrm{Ker}(p^*G \rightarrow_s G))$ functorially in $G \in (\mathrm{l.c.} \mathrm{gr}^{(\mathbf{P})})$;
- (ii) the formation of $\pi_1(X/S, s)^{(\mathbf{P})}$ is compatible with any change of base.

If G and H are two sheaves of groups on S_{et} , we define the sheaf of exterior morphisms from H to G , $\mathrm{Hom}^{\mathrm{ext}}(H, G)$, as the quotient of $\mathrm{Hom}(H, G)$ by the action of H induced by its action on itself by inner automorphism.

The sheaf

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S, s)^{(\mathbf{P})}, G) \simeq R^1p_*(p^*G) \quad (\text{for } G \in (\mathrm{l.c.} \mathrm{gr}^{(\mathbf{P})}))$$

is “independent” of the choice of s .

We shall denote it as

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S)^{(\mathbf{P})}, G).$$

As $p : X \rightarrow S$ admits sections locally for the etale topology, this sheaf makes sense without assuming p to have a global section. It is independent of the choice of $\mathbf{P}$, so long as $\mathbf{P}$ is prime to the order of G and includes all residue characteristics of S .

Definition (5.6). — Let G be a finite group of order n prime to $\mathbf{P}$. A Teichmüller structure of level G on X is a surjective exterior homomorphism from $\pi_1(X/S)$ to G .

The finite generation of $\pi_1(X/S)$ (SGA, 60/61, exp. 10) implies:

Lemma (5.7). — The sheaf on (Sch/S) of the Teichmüller structures of level G on X is represented by an etale covering of S .

We denote by ${}_G\mathcal{M}_g^0$ the stack classifying the stable smooth curves of genus g and characteristic prime to n , with a Teichmüller structure of level G .

For any algebraic stack $\mathcal{M}$, we denote by $\mathcal{M}[1/n]$ its open subset $\mathcal{M} \times \mathrm{Spec}(\mathbb{Z}[1/n])$.

Lemma (5.7) may now be rephrased:

Proposition (5.8). — The “forgetful” morphism

$${}_G\mathcal{M}_g^0 \longrightarrow \mathcal{M}_g^0[1/n]$$

is representable, finite and etale.

The stack ${}_G\mathcal{M}_g^0$ is thus an algebraic stack. Let ${}_G\mathcal{M}_g$ be the normalisation of $\mathcal{M}_g[1/n]$ with respect to ${}_G\mathcal{M}_g^0$. The stack ${}_G\mathcal{M}_g$, being representable and finite over $\mathcal{M}_g[1/n]$, is proper over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Theorem (5.9). — The geometric fibres of the projection of ${}_G\mathcal{M}_g$ onto $\mathrm{Spec}(\mathbb{Z}[1/n])$ are normal, and, fibre by fibre, ${}_G\mathcal{M}_g^0$ is dense in ${}_G\mathcal{M}_g$.

Proof. — We will use Abhyankar–Artin’s lemma, in its “absolute” form:

Lemma (5.10). — *Let D be a divisor with normal crossings on an excellent regular scheme X , Y an étale covering of $X - D$ and $\bar{Y}$ the normalisation of X with respect to Y . Assume that the generic points of the irreducible components of D are all of characteristic 0. Then every geometric point of X has an étale neighbourhood $x : X' \rightarrow X$ such that, on X' :*

- (i) D becomes a union of regular divisors $(D_i)_{i \in I}$, D_i of equation $t_i = 0$.
- (ii) There exists an integer k prime to residue characteristics of X' such that $\bar{Y}$ becomes isomorphic to a disjoint union of quotients (by subgroups of μ_k^I) of the covering of X' obtained by extracting the k^{th} -roots of the t_i ’s.

If $x : X \rightarrow \mathcal{M}_g[1/n]$ is an étale morphism, then $X^0 = x^{-1}(\mathcal{M}_g^0)$ is the complement in X of a divisor with normal crossing relative to $\text{Spec}(\mathbb{Z})$, $X_1^0 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g^0)$ is an étale covering of X^0 and $X_1 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$ is the normalisation of X with respect to this covering of X^0 .

By the explicit local description (5.10), for any prime number l prime to n , $X_1 \times \text{Spec}(\bar{\mathbb{F}}_l)$ is the normalisation of $X \times \text{Spec}(\bar{\mathbb{F}}_l)$ with respect to $X_1^0 \times \text{Spec}(\bar{\mathbb{F}}_l)$. As this is true for any modular family, we get (5.9).

Corollary (5.11). — *The geometric fibres of the projection*

$${}_G\mathcal{M}_g \longrightarrow \text{Spec}(\mathbb{Z}[1/n])$$

all have the same number of connected components.

Proof. — By (4.17), all geometric fibres $\mathcal{M}_g \times \text{Spec}(\bar{\mathbb{F}}_l)$ of $\mathcal{M}_g$ over $\text{Spec}(\mathbb{Z}[1/n])$ have the same number of connected components. These connected components are irreducible (4.16). Furthermore ${}_G\mathcal{M}_g^0 \times \text{Spec}(\bar{\mathbb{F}}_l)$ is dense in ${}_G\mathcal{M}_g \times \text{Spec}(\bar{\mathbb{F}}_l)$ and thus has the same number of connected (or irreducible) components as ${}_G\mathcal{M}_g \times \text{Spec}(\bar{\mathbb{F}}_l)$, and this number is independent of l .

(5.12) Let us denote by Π the fundamental group of an oriented closed differentiable surface S_0 of genus g . The group Π may be defined by generators and relations as follows:

- (i) generators: x_i for $1 \leq i \leq 2g$;
- (ii) relation: $(x_1, x_{g+1}) \cdots (x_i, x_{g+i}) \cdots (x_g, x_{2g}) = e$, where $(a, b) = aba^{-1}b^{-1}$.

Let (e_i) be the standard basis of $\mathbb{Z}^{2g}$. The morphism φ from Π to $\mathbb{Z}^{2g}$ with $\varphi(x_i) = e_i$ identifies $\Pi/(\Pi, \Pi) = H_1(\Pi, \mathbb{Z})$ with $\mathbb{Z}^{2g}$.

The surface S_0 is a $K(\Pi, 1)$ and thus

$$H^2(\Pi, \mathbb{Z}) = H^2(S_0, \mathbb{Z}) = \mathbb{Z},$$

and the cup product defines a symplectic structure on $\Pi/(\Pi, \Pi)$. This structure is identified by φ with the standard symplectic structure of $\mathbb{Z}^{2g}$.

We denote by $\text{Aut}^0(\Pi)$ the subgroup of $\text{Aut}(\Pi)$ which acts trivially on $H^2(\Pi, \mathbb{Z})$. Dehn has proved that each exterior automorphism of Π is induced by a diffeomorphism of S_0 onto itself and that the map induced by φ :

$$\text{Aut}^0(\Pi) \longrightarrow \text{Sp}_{2g}(\mathbb{Z}),$$

is surjective (see [Ma]).

Theorem (5.13). — *The number of connected components of any geometric fibre of the projection of ${}_G\mathcal{M}_g^0$ onto $\text{Spec}(\mathbb{Z}[1/n])$ is equal to the number of orbits of $\text{Aut}^0(\Pi)$ in the set of exterior epimorphisms from Π to G .*

Proof. — By (5.13), it suffices to prove that ${}_G\mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})$ has the said number of connected components. As results from (4.14),

$$\pi_0({}_G\mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})) = \pi_0({}_G\mathcal{M}_g)$$

where ${}_G\mathcal{M}_g$ is the coarse moduli scheme classifying stable smooth curves of genus g over $\mathbb{C}$ with a Teichmüller structure of level G .

Recall that a Teichmüller curve of genus g is a stable smooth curve C of genus g over $\mathbb{C}$ together with an exterior isomorphism φ of the transcendental fundamental group $\pi_1(C)$ with Π , which induces a symplectic isomorphism¹ between

$$H_1(C, \mathbb{Z}) \simeq \pi_1(C)/(\pi_1(C), \pi_1(C))$$

¹An arbitrary isomorphism φ induces an isomorphism between $H_1(C, \mathbb{Z})$ and $\Pi/(\Pi, \Pi)$ which is always symplectic up to sign.

and $\Pi/(\Pi, \Pi)$. By Teichmüller's theory [W], the analytic space T_g classifying Teichmüller curves of a given genus $g \geq 2$ is homeomorphic to a ball, and hence connected.

If ψ is a surjective homomorphism from Π to G , the map

$$(C, \varphi) \mapsto (C, \psi\varphi)$$

defines a morphism $t_\psi : T_g \rightarrow {}_G M_g$. Two such maps t_ψ and $t_{\psi'}$ have the same image if and only if $\psi = \psi'\sigma$ for $\sigma \in \text{Aut}^0(\Pi)$, and

$${}_G M_g = \coprod_{\psi \bmod \text{Aut}^0(\Pi)} t_\psi(T_g)$$

which implies (5.13).

(5.14) Let us denote by ${}_n \mathcal{M}_g^0$ the algebraic stack classifying stable smooth curves with a Jacobi structure of level n . This algebraic stack “is” a true scheme for $n \geq 3$ (by [S]).

If φ is a Jacobi structure of level n on a stable smooth curve $p : X \rightarrow S$, we define the “multiplier” $\mu(\varphi)$ of φ by the commutative diagram

$$\begin{array}{ccc} \Lambda^2(\mathbb{Z}/n\mathbb{Z})^{2g} & \xrightarrow{A} & \mathbb{Z}/n\mathbb{Z} \\ \Lambda^2\varphi \uparrow & & \uparrow \mu(\varphi) \\ \Lambda^2 R^1 p_*(\mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\Lambda} & \mu_n^{\otimes -1} \simeq R^2 p_*(\mathbb{Z}/n\mathbb{Z}) \end{array}$$

The scheme of isomorphisms between $\mathbb{Z}/n\mathbb{Z}$ and $\mu_n^{\otimes -1}$ is $\text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$; thus $\varphi \mapsto \mu(\varphi)$ defines a morphism μ from S to $\text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This being true for any X and S , μ is induced by

$$\mu : {}_n \mathcal{M}_g^0 \longrightarrow \text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n]).$$

Theorem (5.15). — *The geometric fibres of the morphism μ are connected.*

Proof. — By definition, ${}_n \mathcal{M}_g^0$ is open and closed in ${}_G \mathcal{M}_g^0$ for $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$. The group $\text{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$ acts on G , and thus on ${}_G \mathcal{M}_g^0$. One has:

- (i) the open subset ${}_n \mathcal{M}_g^0$ of ${}_G \mathcal{M}_g^0$ is stable under the subgroup $H = \text{CSp}_{2g}(\mathbb{Z}/n\mathbb{Z})$ of symplectic similitudes;
- (ii) ${}_G \mathcal{M}_g^0 = \coprod_{\sigma/H} \sigma({}_n \mathcal{M}_g^0)$.

It now results from (5.11) that all geometric fibres of μ have the same number of connected components.

Consider the geometric fibre of μ at the standard complex place of $\text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This fibre is the algebraic stack classifying complex stable smooth curves C provided with a symplectic isomorphism

$$H^1(C, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^{2g}.$$

Reasoning as in (5.13), we are reduced to proving

Lemma (5.16). — *The homomorphism*

$$\text{Aut}^0(\Pi) \longrightarrow \text{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Proof. — This results from Dehn's theory (5.12) and from the fact that the groups Sp_{2g} , being split semi-simple simply connected groups, are generated by their unipotent elements, so that the reduction map

$$\text{Sp}_{2g}(\mathbb{Z}) \longrightarrow \text{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

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